

ASSESSMENT TOOL 3: Class Test

Course Outcome Covered

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BT L3)

Assessment Tool Weightage: 10%

Course Name: Computer Vision with OpenCV

Course Code: DSA0204

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1. An image appears too bright. Explain and justify the most suitable grey level transformation to improve visibility, clearly describing the concept, application process, and expected outcome

Suitable Technique: Negative Transformation

* Concept: Negative transformation reverses the gray level of an image.

* Formula: $S = L - 1 - Y$

* Application Process:

1. Read the input image.
2. Apply the negative transformation to each pixel.
3. Display the transformed image.

* Expected Outcome:

→ Reduces excessive brightness.

→ Improves visibility of bright regions.

→ Enhances hidden details.

2. A dark image lacks detail. Describe and justify an appropriate grey level transformation to enhance visibility, explaining how it improves low-intensity regions with proper reasoning.

Suitable Technique: Log Transformation

* Concept: Log transformation expands dark pixel values and compresses brighter values.

* Formula: $s = c \log(1+r)$

* Application Process:

1. Read the image.
2. Apply log transformation.
3. Normalize intensity values.

* Expected outcome:

- Brightens dark areas.
- Reveals hidden details.
- Improves overall visibility.

3. An image has poor contrast. Explain a suitable histogram based technique for contrast enhancement and illustrate how it improves image quality with logical reasoning.

Suitable Technique: Histogram Equalization

* Concept: Redistributes image intensity values uniformly.

* Application Process:

1. Compute the histogram.
2. Calculate the cumulative distribution function (CDF).
3. Map old intensity values to new values.

* Expected Outcome:

- Enhances image contrast.

- improves visibility of objects.
- Produces a clearer image.

4. An image shows uneven brightness across regions. Explain and apply an appropriate enhancement technique to correct non-uniform illumination, providing a structured explanation.

Suitable Technique: Adaptive Histogram Equalization (CLAHE)

* Concept: improves local contrast by processing small image regions separately.

* Application Process:

1. Divide the image into small blocks.
2. Apply histogram equalization to each block.
3. Merge the blocks smoothly.

* Expected Outcome:

- corrects uneven illumination.
- Preserves local details.
- Produces balanced brightness.

5. A noisy image contains random intensity variations. Explain and justify the most suitable smoothing filter, including its working mechanism and impact on noise reduction.

Suitable Technique: Mean Filter

* Concept: Replaces each Pixel with the average of neighboring pixels.

* Application Process:

1. Select a 3×3 filter window.
2. Compute the average of neighboring pixels.
3. Replace the center pixel.

* Expected Outcome:

- Reduces random noise.
- Smooths the image.
- Slightly blurs edges.

6. An image appears blurred after smoothing. Explain and Justify a sharpening method to restore details, including its working principle and expected results.

Suitable Technique: Unsharp Masking

* Concept: Enhances image sharpness by adding high-frequency details.

* Application Process:

1. Blur the original image.
2. Subtract the blurred image from the original.
3. Add the difference back to the original.

* Expected Outcome:

- Restores sharpness.
- Enhances edges.
- Improves image clarity.

7. A noisy image must be preprocessed before further analysis. Explain and apply a suitable spatial filtering method, ensuring a clear, structured explanation with justification.

Suitable Technique: Gaussian filter

* Concept: Smooths the image using a Gaussian kernel while preserving overall structure.

* Application Process:

1. Create a Gaussian kernel.
2. Convolve the kernel with the image.
3. Generate the filtered image.

* Expected outcome:

- Reduces Gaussian noise.

→ Produces smooth images.

→ Minimizes distortion.

8. An object is clearly brighter than the background. Explain and justify the segmentation technique used, including its working and effectiveness.

Suitable Technique: Threshold Segmentation

* Concept: Separates object and background using a threshold value.

* Application Process:

1. Select threshold T .
2. Pixels greater than T become object pixels.
3. Remaining pixels become background.

* Expected outcome:

- Accurate object extraction.
- Simple and efficient segmentation.

9. An image requires identification of object edges. Explain and justify a suitable edge detection method, describing its working and effectiveness.

Suitable Technique: Canny Edge Detection

* Concept: Detects edges using gradients information with noise suppression.

* Application Process:

1. Smooth the image.
2. Compute image gradients.
3. Apply non-maximum suppression.
4. Use double thresholding and edge tracking.

Expected Outcome:

- Detects accurate edges.

→ Produces continuous boundaries.

→ Reduces false edge detection.

10. An image contains salt-and-pepper noise. Explain and justify the most appropriate spatial filter, describing its working mechanism and effectiveness in noise removal.

Suitable Technique: Median Filter

* Concept: Replaces each Pixel with the median of neighboring Pixel values.

* Application Process:

1. Select a 3×3 neighborhood.

2. Arrange pixel values in ascending order.

3. Replace the center pixel with the median value.

* Expected Outcome:

→ Removes salt-and-pepper noise effectively.

→ Preserves image edges.

→ Improves image quality.